Linear algebra – new problems

Problems built on recent research results

Each problem is a special case of a published result or of a lemma used in its proof; the papers are in this folder. The difficulty is the position of the problem in an IMC day: 1 easy, 2 below average, 3 medium, 4 above average, 5 hard.

Problem 1. *IMC level: 1/5 (easy)*

For an $n \times n$ matrix $A = (a_{ij})$ with entries in $\{0,1\}$ and a vector $x \in \{0,1\}^n$ define the vector $A \odot x \in \{0,1\}^n$ by

$$(A \odot x)_i = \max_{1 \leqslant j \leqslant n} a_{ij} x_j, \qquad i = 1, \ldots, n.$$

Suppose that the map $x \mapsto A \odot x$ from $\{0,1\}^n$ to itself is injective. Prove that A is a permutation matrix. Deduce that the map $x \mapsto A^{\mathsf{T}} \odot x$ is injective as well.

Based on: S. Gu, W. Qi, Y. Cheng, *Transpose symmetry of injectivity over commutative semirings*, arXiv:2608.16205, Theorem 1 (the map $x \mapsto Ax$ is injective if and only if $x \mapsto A^{\mathsf{T}}x$ is, over every commutative semiring); here the semiring is $(\{0,1\}, \max, \min)$.

Problem 2. *IMC level: 2/5 (below average)*

Call a set $\{a,b,c,d\}$ of four distinct positive integers *spectral* if each of the 24 matrices $\begin{pmatrix} p & q \\ r & s \end{pmatrix}$, where (p,q,r,s) is a permutation of (a,b,c,d), has integer eigenvalues.

(a) Show that $\{1,6,7,12\}$ is spectral.

(b) Prove that there are infinitely many spectral sets whose elements have no common divisor greater than 1.

Based on: M. J. W. Hall, *Matrices with integer eigenvalues for all permutations of coefficients (thanks to Pythagoras!)*, arXiv:2601.01083, equations (7)–(17): spectral sets obtained from Pythagorean triples.

Problem 3. *IMC level: 3/5 (medium)*

Let $k, m, r_1, \ldots, r_m$ be positive integers. The tree T is obtained from a vertex v_0 by attaching m paths: the i-th path has r_i vertices, and one of its end vertices is joined to v_0. Let B be the symmetric matrix whose rows and columns are indexed by the vertices of T, with

$$B_{v_0v_0} = k, \qquad B_{vv} = 2 \ \ (v \neq v_0), \qquad B_{uv} = \begin{cases} -1, & u, v \text{ adjacent}, \\ 0, & \text{otherwise} \end{cases} \qquad (u \neq v).$$

(a) Prove that $\det B = (r_1+1)\cdots(r_m+1)\Big(k - \displaystyle\sum_{i=1}^{m} \frac{r_i}{r_i+1}\Big)$.

(b) Prove that B is positive definite if and only if $\displaystyle\sum_{i=1}^{m} \frac{1}{r_i+1} > m - k$.

(c) Let $k = 2$ and $m = 3$. Find all triples $r_1 \leqslant r_2 \leqslant r_3$ for which $\det B = 0$.

Based on: E. Torrente-Lujan, *Star-shaped integral Cartan-type matrices and an Egyptian-fraction classification of affine weighted trees*, arXiv:2605.23011, Proposition 4.1, Theorem 4.2 and Corollary 4.3.

Problem 4. *IMC level: 4/5 (above average)*

Let A be a complex 2×2 matrix with eigenvalues λ and μ (counted with multiplicity). Prove that the set

$$W(A) = \{\, u^* A u \;:\; u \in \mathbb{C}^2,\ u^* u = 1 \,\}$$

is the closed region bounded by an ellipse (possibly degenerating into a segment) with foci λ and μ whose minor axis has length $\sqrt{\operatorname{tr}(A^* A) - |\lambda|^2 - |\mu|^2}$.

Based on: P. Blaschke, *Coval description of the boundary of a numerical range and the secondary values of a matrix*, arXiv:2410.03744, Proposition 1 (the elliptical range theorem of Toeplitz).

Problem 5. *IMC level: 5/5 (hard)*

Let A be a complex $n \times n$ matrix. Prove that $A = B + C$ for some complex $n \times n$ matrices B and C with $B^2 = C^2 = 0$ if and only if A is similar to $-A$.

Based on: C. de Seguins Pazzis, *Sums of two square-zero selfadjoint or skew-selfadjoint endomorphisms*, arXiv:2210.03955, Theorems 1.1 and 1.2 (the theorem of Wang–Wu and Botha).